Umesh Kumar Sahoo

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Profile Summary

Computer Science undergraduate with hands-on experience building web applications, ML APIs, and RAG-based AI systems. Skilled in Python, Basic JavaScript, SQL, REST APIs, machine learning, embeddings, vector search, and LLM integration, with a strong interest in software development and AI engineering.

Education

B.Tech in Computer Science & Engineering
Silicon University, Bhubaneswar 2023–2027
CGPA: 9.60 / 10.0

Senior Secondary
JNV Panikoili 2021
Percentage: 91.2%

Secondary
JNV Panikoili 2019
Percentage: 95.8%

Technical Skills

Programming: Python, C, SQL
Machine Learning / AI: Scikit-learn, XGBoost, Gradient Boosting, Random Forest, Keras, RAG Pipelines, LLM Integration, Embeddings, Vector Similarity Search, Feature Engineering, Model Evaluation
Frameworks / Libraries: FastAPI, Flask, React.js, Node.js, Pandas, NumPy, Matplotlib, Seaborn
Tools / DevOps: Git, GitHub, GitHub Actions, Terraform, Linux, Postman, Jupyter Notebook, VS Code

Professional Experience

DevOps Engineer Intern, *Ingenious Tech World* *June 2025 – July 2025*

- Worked with GitHub Actions and Terraform on CI/CD and Infrastructure-as-Code workflows.

MERN Stack Developer Intern, *Syllogistick* *June 2024 – July 2024*

- Worked on a MERN-based application involving React, REST APIs, authentication, and MongoDB.

Projects

AI-Powered Terminal Portfolio

React •FastApi• Gemini • REST APIs • Security

- Built a React + FastApi terminal-style portfolio with an backend, integrating GitHub and Hacker News APIs with server-side caching for dynamic content.
- Developed a Gemini-powered assistant using resume and public repository context to answer questions about projects, skills, and experience.
- Implemented API security controls including Helmet security headers, rate limiting, CSRF protection, input validation, and server-side secret management.

RAG-Based Teaching Assistant System

Python • RAG • Embeddings • Vector Search • LLM Integration

- Built an end-to-end RAG pipeline covering document ingestion, chunking, embedding generation, vector similarity search, retrieval, and LLM-based response generation.
- Used semantic retrieval to provide relevant source context to the language model for context-aware question answering over educational content.

Emotion Detection API

Python • Keras • BiGRU • FastAPI • NLP

- Built a Keras Bidirectional GRU model for text emotion classification and exposed inference through a FastAPI REST service.
- Implemented inference preprocessing, tokenizer-based sequence conversion, padding, and prediction with emotion probabilities and confidence scores.
- Added health and prediction endpoints with validation and error handling for unavailable model or tokenizer artifacts.

Achievements

- **Runner-Up – SPEC2MODEL ML Challenge:** Secured 2nd place in the Food Delivery Delay Prediction competition.
- **Algo Arena 2.0:** Secured 2nd Place at Silicon University.
- **PwC Launchpad:** Selected for the PwC Launchpad program.
- **Kaggle Agents Intensive Capstone:** Participated in the Google x Kaggle Agents Intensive program and built an agent-based AI application using Google ADK and Gemini.

Certifications

Google – Generative AI Introduction • IBM – Introduction to Artificial Intelligence • CodeWithHarry – Job Ready Data Science